(i)

$$P_2 = \frac{1}{2}$$

$$T_1 \text{ sits in } S_1 : \ T_3 \text{ can sits in } S_3$$

$$T_1 \text{ sits in } S_2 : \ T_2 \text{ chooses } S_1 \text{ or } S_3$$

$$T_1 \text{ sits in } S_3 : \ T_3 \text{ cannot sit in } S_3$$

$$P_3 = \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$$

(ii)

$$T_1 \text{ sits in } S_1 : \quad P(T_n \text{ sits in } S_n) = 1$$
$$T_1 \text{ sits in } S_n : \quad P(T_n \text{ sits in } S_n) = 0$$

If T_1 sits in S_k, where $2 \leq k \leq n-1$,
If T_1 sits in S_k, where $2 \leq k \leq n-1$, $T_2, T_3, \ldots, T_{k-1}$ will sit in their own seats
Then T_k is the first passenger whose seat is not available
There are $n-k+1$ available seats left: $S_1, S_{k+1}, S_{k+2}, \ldots, S_n$

$$P(T_n \text{ sits in } S_n \mid T_1 \text{ sits in } S_k) = P_{n-k+1}$$

This becomes the same problem with $n-k+1$ seats.

$$\begin{aligned} P_n &= \frac{1}{n} \cdot 1 + \frac{1}{n} \sum_{k=2}^{n-1} P_{n-k+1} + \frac{1}{n} \cdot 0 \\ &= \frac{1}{n} \left(1 + \sum_{k=2}^{n-1} P_{n-k+1} \right) \\ &= \frac{1}{n} \left(1 + \sum_{r=2}^{n-1} P_r \right) \end{aligned}$$

(iii)

Assume $P_k = \frac{1}{2} \quad (k \geq 2)$

Base case: $P_2 = \frac{1}{2}$

$$P_k = \frac{1}{k}\left(1 + \sum_{r=2}^{k-1} P_r\right) \implies 1 + \sum_{r=2}^{k-1} P_r = kP_k$$

$$\begin{aligned} P_{k+1} &= \frac{1}{k+1}\left(1 + \sum_{r=2}^{k} P_r\right) \\ &= \frac{1}{k+1}\left(1 + \sum_{r=2}^{k-1} P_r + P_k\right) \\ &= \frac{1}{k+1}(kP_k + P_k) \\ &= \frac{1}{2} \end{aligned}$$

By mathematical induction, $P_n = \frac{1}{2}$

(iv)

$$Q_2 = \frac{1}{2}$$

For $n \geq 3$, condition on the seat chosen by T_1.

$$T_1 \text{ sits in } S_1 \text{ or } S_n : \ T_{n-1} \text{ sits in } S_{n-1}$$

$$T_1 \text{ sits in } S_{n-1} : \ T_{n-1} \text{ cannot sit in } S_{n-1}$$

If T_1 sits in S_k, where $2 \leq k \leq n-2$,

$$T_k \text{ becomes the first displaced passenger}$$

$$P(T_{n-1} \text{ sits in } S_{n-1} \mid T_1 \text{ sits in } S_k) = Q_{n-k+1}$$

$$\begin{aligned} Q_n &= \frac{1}{n}\left(2 + \sum_{k=2}^{n-2} Q_{n-k+1}\right) \\ &= \frac{1}{n}\left(2 + \sum_{r=3}^{n-1} Q_r\right) \end{aligned}$$

Base case: $Q_3 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$

Assume $Q_3 = Q_4 = \cdots = Q_{n-1} = \frac{2}{3}$

$$
\begin{aligned}
Q_n &= \frac{1}{n}\left(2 + \sum_{r=3}^{n-1} Q_r\right) \\
&= \frac{1}{n}\left(2 + \frac{2(n-3)}{3}\right) \\
&= \frac{2}{3}
\end{aligned}
$$

$$
\therefore Q_n = \begin{cases} \frac{1}{2}, & n = 2, \\ \frac{2}{3}, & n \geq 3. \end{cases}
$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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